

```
<!DOCTYPE html>
<html lang="ru">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, viewport-fit=cover">
  <title>Weather Gate v5.3 PRO</title>
  <!-- Локальные библиотеки из LittleFS памяти платы ESP32 -->
  <link rel="stylesheet" href="/uPlot.min.css">
  <script src="/sha256.js"></script>
  <style>
    * { margin: 0; padding: 0; box-sizing: border-box; }
    :root {
      --bg: #f5f5f7;
      --card: #ffffff;
      --text: #1e1e1e;
      --text-secondary: #6c6c74;
      --temp: #ff6b35;
      --humidity: #2196f3;
      --pressure: #4caf50;
      --wind: #9c27b0;
      --rain: #00bcd4;
      --success: #4caf50;
      --error: #f44336;
      --warning: #ff9800;
      --radius: 24px;
      --radius-sm: 16px;
      --shadow: 0 2px 8px rgba(0,0,0,0.05);
      --forecast: #ff6b35;
    }
    body.dark {
      --bg: #121212;
      --card: #1e1e1e;
      --text: #e3e3e3;
      --text-secondary: #9e9e9e;
    }
    body {
      font-family: system-ui, -apple-system, sans-serif;
      background: var(--bg);
      padding: 12px;
      color: var(--text);
      transition: background 0.3s, color 0.3s;
    }
    .container { max-width: 550px; margin: 0 auto; display: flex; flex-direction: column; gap: 12px; }
    .cards-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 12px; }
    .m-card { background: var(--card); border-radius: var(--radius); padding: 14px; box-shadow: var(--shadow); position: relative; }
  </style>
  .m-card.wide { grid-column: span 2; }

  .card-header { display: flex; align-items: center; gap: 12px; margin-bottom: 8px; }
  .card-icon { width: 44px; height: 44px; border-radius: 50%; display: flex; align-items: center; justify-content: center; font-size: 22px; }
  .card-info { flex: 1; }
  .card-label { font-size: 12px; font-weight: 600; color: var(--text-secondary); text-transform: uppercase; letter-spacing: 0.5px; }
  .card-value { font-size: 28px; font-weight: 800; line-height: 1.2; }
  .card-sub { font-size: 12px; font-weight: 500; color: var(--text-secondary); margin-top: 4px; }

  .status-row { display: flex; justify-content: space-between; margin-top: 12px; font-size: 13px; font-weight: 500; color: var(--text-secondary); }
  .status-row b { color: var(--text); font-weight: 700; }
  .badge { display: inline-block; padding: 4px 8px; border-radius: 12px; font-size: 11px; font-weight: 700; }
```

```

.badge-success { background: rgba(76,175,80,0.15); color: var(--success); }
.badge-error { background: rgba(244,67,54,0.15); color: var(--error); }
.badge-warning { background: rgba(255,152,0,0.15); color: var(--warning); }

/* Роза ветров тригонометрического усреднения */
.compass { position: relative; width: 140px; height: 140px; margin: 10px auto; }
.compass-bg { width: 140px; height: 140px; background: var(--bg); border-radius: 50%; position: relative; background-
image: radial-gradient(circle at center, transparent 50%, rgba(0,0,0,0.04) 100%); }
.compass-arrow { position: absolute; top: 50%; left: 50%; width: 4px; height: 55px; background: var(--wind); transform-
origin: 50% 0%; transition: transform 0.5s ease; z-index: 2; }
.compass-arrow::after { content: "▼"; position: absolute; bottom: -10px; left: -6px; font-size: 16px; color: var(--wind); }
.compass-label { text-align: center; margin-top: 8px; font-size: 14px; font-weight: bold; color: var(--wind); }
.compass-mark { position: absolute; font-size: 10px; font-weight: 600; color: var(--text-secondary); transform: translate(-
50%, -50%); }

.forecast-card { background: var(--card); border-radius: var(--radius); padding: 14px; box-shadow: var(--shadow); text-
align: center; grid-column: span 2; }
.forecast-text { font-size: 22px; font-weight: 800; color: var(--forecast); margin: 8px 0; }
.forecast-icon { font-size: 40px; margin-bottom: 6px; }

.power-progress { margin-top: 12px; }
.power-labels { display: flex; justify-content: space-between; font-size: 10px; font-weight: 500; color: var(--text-
secondary); margin-bottom: 4px; }
.progress-bar { width: 100%; height: 6px; background: rgba(0,0,0,0.08); border-radius: 3px; overflow: hidden; }
body.dark .progress-bar { background: rgba(255,255,255,0.08); }
.progress-fill { height: 100%; width: 0%; transition: width 0.3s; border-radius: 3px; }
.chart-container { margin-top: 12px; width: 100%; height: 140px; }

#spectrumCanvas { cursor: crosshair; transition: background 0.2s; border-radius: 16px; width: 100%; height: 180px;
background: var(--bg); }

.settings-panel { background: var(--card); border-radius: var(--radius); margin-top: 8px; box-shadow: var(--shadow); }
.settings-header { padding: 16px; display: flex; justify-content: space-between; align-items: center; cursor: pointer; font-
weight: 700; }
.settings-content { max-height: 0; overflow: hidden; transition: max-height 0.3s ease-out; padding: 0 16px; }
.settings-content.open { max-height: 2000px; padding: 0 16px 16px 16px; }
.settings-row { display: flex; gap: 12px; margin-bottom: 12px; flex-wrap: wrap; }
.settings-row > div { flex: 1; min-width: 120px; }

input { width: 100%; padding: 10px 12px; border-radius: 12px; border: none; background: var(--bg); color: var(--text);
font-size: 14px; font-weight: 500; outline: none; }
body.dark input { background: #2c2c2e; }
button { width: 100%; padding: 12px; border-radius: 12px; border: none; background: var(--temp); color: white; font-
weight: 700; cursor: pointer; transition: transform 0.1s; }

```

```

button:active { transform: scale(0.98); }
button.secondary { background: var(--text-secondary); }
button.success { background: var(--pressure); }
button.warning { background: var(--warning); color: #1e1e1e; }
.btn-grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 10px; margin-top: 4px; }

```

```

@keyframes slideUp { from { transform: translateY(15px); opacity: 0; } to { transform: translateY(0); opacity: 1; } }

```

```

</style>

```

```

</head>

```

```

<body class="dark">

```

```

<div class="container">

```

```

<!-- Первый ряд: Температура и Влажность (Fine Offset WH1080) -->

```

```

<div class="cards-grid">

```

```

  <div class="m-card">

```

```

    <div class="card-header">

```

```

      <div class="card-icon" style="background:rgba(255,107,53,0.1); color: var(--temp);"> 🌡️ </div>

```

```

      <div class="card-info">

```

```

        <div class="card-label">Температура</div>

```

```

        <div class="card-value"><span id="temperature">---</span> °C</div>

```

```

      </div>

```

```

    </div>

```

```

    <div class="status-row">

```

```

      <span>Мин: <b><span id="temp_min">---</span> °C</b></span>

```

```

      <span>Макс: <b><span id="temp_max">---</span> °C</b></span>

```

```

    </div>

```

```

    <div class="power-progress">

```

```

      <div class="power-labels"><span>-30°C</span><span>0°C</span><span>+30°C</span><span>+50°C</span></div>

```

```

      <div class="progress-bar"><div id="temp-fill" class="progress-fill" style="background: var(--temp);"></div></div>

```

```

    </div>

```

```

  </div>

```

```

<div class="m-card">

```

```

  <div class="card-header">

```

```

    <div class="card-icon" style="background:rgba(33,150,243,0.1); color: var(--humidity);"> 💧 </div>

```

```

    <div class="card-info">

```

```

      <div class="card-label">Влажность</div>

```

```

      <div class="card-value"><span id="humidity">---</span> %</div>

```

```

    </div>

```

```

  </div>

```

```

  <div class="power-progress" style="margin-top: 31px;">

```

```

    <div class="power-labels"><span>0%</span><span>25%</span><span>50%</span><span>75%</span>

```

```

    <span>100%</span></div>

```

```

    <div class="progress-bar"><div id="hum-fill" class="progress-fill" style="background: var(--humidity);"></div></div>

```

```
    </div>
  </div>
</div>
```

```
<!-- Второй ряд: Давление (BMP280) и Осадки (WH1080) -->
```

```
<div class="cards-grid">
```

```
  <div class="m-card">
```

```
    <div class="card-header">
```

```
      <div class="card-icon" style="background:rgba(76,175,80,0.1); color: var(--pressure);"><img alt="Bar chart icon" data-bbox="738 245 758 258"/></div>
```

```
      <div class="card-info">
```

```
        <div class="card-label">Давление</div>
```

```
        <div class="card-value" style="font-size:24px;"><span id="pressure">--</span> hPa</div>
```

```
      </div>
```

```
    </div>
```

```
    <div class="status-row">
```

```
      <span>Мин: <b><span id="pressure_min">--</span></b></span></div>
```

```
      <span>Макс: <b><span id="pressure_max">--</span></b></span></div>
```

```
    </div>
```

```
    <div class="power-progress">
```

```
      <div class="power-labels"><span>970</span><span>1013</span><span>1050</span></div>
```

```
      <div class="progress-bar"><div id="pressure-fill" class="progress-fill" style="background:
```

```
var(--pressure);"></div></div>
```

```
    </div>
```

```
  </div>
```

```
<div class="m-card">
```

```
  <div class="card-header">
```

```
    <div class="card-icon" style="background:rgba(0,188,212,0.1); color: var(--rain);"><img alt="Rain cloud icon" data-bbox="703 597 723 610"/></div>
```

```
    <div class="card-info">
```

```
      <div class="card-label">Осадки всего</div>
```

```
      <div class="card-value"><span id="rain">--</span> mm</div>
```

```
    </div>
```

```
  </div>
```

```
  <div class="status-row">
```

```
    <span>Макс/Импульс: <b><span id="rain_max">--</span> mm</b></span></div>
```

```
  </div>
```

```
  <div class="power-progress">
```

```
    <div class="power-labels"><span>0</span><span>10</span><span>25</span><span>50</span><span>100+</
```

```
span></div>
```

```
    <div class="progress-bar"><div id="rain-fill" class="progress-fill" style="background: var(--rain);"></div></div>
```

```
  </div>
```

```
</div>
```

```
<!-- Третий ряд: Ветер (скорость) и Направление (роза ветров) -->
```

```

<div class="cards-grid">
  <div class="m-card">
    <div class="card-header">
      <div class="card-icon" style="background:rgba(156,39,176,0.1); color: var(--wind);">🌬️</div>
      <div class="card-info">
        <div class="card-label">Ветер</div>
        <div class="card-value"><span id="wind_speed">--</span> м/с</div>
        <div class="card-sub">Порывы: <span id="wind_gust">--</span> м/с</div>
      </div>
    </div>
    <div class="status-row">
      <span>Макс: <b><span id="wind_max">--</span> м/с</b></span>
    </div>
    <div class="power-progress">
      <div class="power-labels"><span>0</span><span>10</span><span>20</span><span>30</span><span>40</span>
      м/с</div>
      <div class="progress-bar"><div id="wind-fill" class="progress-fill" style="background: var(--wind);"></div></div>
    </div>
  </div>

  <div class="m-card">
    <div class="card-header">
      <div class="card-icon" style="background:rgba(156,39,176,0.1); color: var(--wind);">🌪️</div>
      <div class="card-info">
        <div class="card-label">Направление</div>
        <div class="card-value" id="wind_dir_deg" style="font-size: 20px;">--°</div>
        <div class="card-sub" id="wind_dir_txt">--</div>
      </div>
    </div>
    <div class="compass" id="compass">
      <div class="compass-bg" id="compassBg"></div>
      <div id="windArrow" class="compass-arrow"></div>
      <div id="windDirLabel" class="compass-label">--</div>
    </div>
  </div>

<!-- Прогноз Zambretti (широкий) -->
<div class="forecast-card">
  <div class="card-header">
    <div class="card-icon" style="background:rgba(255,107,53,0.1); color: var(--forecast);">🌤️</div>
    <div class="card-info">
      <div class="card-label">Прогноз (Zambretti)</div>
      <div class="forecast-text" id="forecast_text">--</div>
    </div>
  </div>

```

```

    <div class="card-sub">на ближайшие 12-24 часа</div>
  </div>
</div>
<div id="forecast_icon" class="forecast-icon"> ⌚ </div>
<div class="status-row" style="justify-content: center;">
  <span>Тенденция: <b><span id="pressure_trend">---</span></b></span></div>
</div>
</div>

```

```

<!-- График температуры uPlot -->
<div class="m-card wide">
  <div class="card-header">
    <div class="card-icon" style="background:rgba(255,107,53,0.1); color: var(--temp);"> 📈 </div>
    <div class="card-info">
      <div class="card-label">Температура</div>
      <div class="card-sub">последние 24 часа (5-минутные интервалы)</div>
    </div>
  </div>
  <div id="temp-chart" class="chart-container"></div>
</div>

```

```

<!-- График давления uPlot -->
<div class="m-card wide">
  <div class="cards-grid" style="display:block;">
    <div class="card-header">
      <div class="card-icon" style="background:rgba(76,175,80,0.1); color: var(--pressure);"> 📈 </div>
      <div class="card-info">
        <div class="card-label">Давление</div>
        <div class="card-sub">последние 24 часа (5-минутные интервалы)</div>
      </div>
    </div>
  </div>
  <div id="pressure-chart" class="chart-container"></div>
</div>

```

```

<!-- Спектроанализатор радиоканала CC1101 -->
<div class="m-card wide">
  <div class="card-header">
    <div class="card-icon" style="background:rgba(33,150,243,0.1); color: #2196F3;"> 📶 </div>
    <div class="card-info">
      <div class="card-label">АЧХ радиоканала гетеродина</div>
      <div class="card-sub">диапазон 914.80 - 915.20 MHz</div>
    </div>
  </div>
</div>

```

```

<canvas id="spectrumCanvas" width="600" height="180"></canvas>
  <div style="display: flex; justify-content: space-between; align-items: center; margin-top: 10px; flex-wrap: wrap; gap:
8px;">
    <div style="display: flex; gap: 16px;">
      <span style="font-size: 13px; font-weight: 600; font-family: monospace;" id="current_freq">915.00</span>
      <span style="font-size: 13px;">MHz</span>
      <span style="font-size: 13px;">📶 <b id="radio_success_header">0</b></span>
      <span style="font-size: 13px;">⚠️ <b id="radio_errors_header">0</b></span>
      <span style="font-size: 13px;">📶 <span id="scan_rssi">---</span> dBm</span>
    </div>
    <!-- Инженерная кнопка сканера: скрывается по умолчанию для роли user -->
    <button id="startScanBtn" onclick="startFullScan()" style="width: auto; padding: 8px 20px; background: var(--success);
display: none;">🔍 Найти лучшую частоту</button>
  </div>
  <div class="status-row" style="margin-top: 8px; font-size: 11px;" id="spectrum_legend_block">
    <span>🟦 Синяя линия — уровень сигнала</span>
    <span>🔴 Пунктир — порог шума</span>
    <span>🟡 Маркер — текущая частота</span>
  </div>
</div>

<!-- Диагностика Системы (Доступна для чтения всем) -->
<div class="m-card wide">
  <div class="card-header">
    <div class="card-icon" style="background: rgba(76,175,80,0.1); color: var(--success);">⚙️</div>
    <div class="card-info">
      <div class="card-label">Система и диагностика связи</div>
    </div>
  </div>
  <div class="status-row">
    <span>Питание: <b><span id="power_source">AC</span></b></span>
    <span>Сеть MQTT: <b><span id="mqtt_status">---</span></b></span>
    <span>Версия ПО: <b><span id="version">5.3</span></b></span>
  </div>
  <div class="status-row">
    <span>Uptime: <span id="uptime">0</span></span>
    <span>Радио: <b><span id="radio_status">OK</span></b></span>
    <span>ID датчика: <b><span id="sensor_id">---</span></b></span>
  </div>
  <div class="status-row">
    <span>Батарея WH1080: <b><span id="outdoor_battery">---</span></b></span>
    <span>Внутренняя плата BMP280 T: <b><span id="bmp_temp_val">0.0</span> °C</b></span>
    <span><span id="datetime">---</span></span>
  </div>
</div>

```

</div>

<!-- Русифицированная плитка базовых кнопок -->

<div class="m-card wide">

<div class="btn-grid">

<button class="secondary" onclick="syncTime()">Синхр. время 🕒</button>

<button onclick="toggleTheme()" class="secondary">🌑 Тёмная тема</button>

</div>

</div>

<!-- СКРЫТАЯ ИНЖЕНЕРНАЯ ПАНЕЛЬ НАСТРОЕК (ОТКРЫВАЕТСЯ СТРОГО ПОСЛЕ ВВОДА PIN) -->

<div class="settings-panel" id="settingsPanelBlock" style="display: none;">

<div class="settings-header" onclick="toggleSettings()">

ИНЖЕНЕРНЫЕ НАСТРОЙКИ МЕТЕОШЛЮЗА ⚙️

▼

</div>

<div id="settingsContent" class="settings-content">

<div class="settings-row">

<div><label>Имя устройства (Hostname) 🏠</label><input type="text" id="hostname"></div>

<div><label>Адрес MQTT Брокера 🌐</label><input type="text" id="mq_ip"></div>

</div>

<div class="settings-row">

<div><label>Логин MQTT 👤</label><input type="text" id="mq_u"></div>

<div><label>Пароль MQTT 🔒</label><input type="password" id="mq_p"></div>

</div>

<div class="settings-row">

<div><label>MQTT топик данных шлюза 📊</label><input type="text" id="mqtt_topic_weather"></div>

<div><label>Высота над уровнем моря (метры) 🏔️</label><input type="number" id="altitude_input"></div>

</div>

<div class="settings-row">

<div><label>Смещение барометра BMP280 (hPa) 📏</label><input type="text" id="bmp_offset"></div>

<div><label>Новый инженерный PIN-код ADMIN 🔑</label><input type="password" id="web_password"

placeholder="Оставьте пустым"></div>

</div>

<div class="settings-row" style="display: flex; gap: 10px;">

<button onclick="resetRadioErrors()" class="warning" style="flex: 1;">Сбросить сбои радио 📶</button>

<button onclick="save()" class="success" style="flex: 1;">Сохранить калибровки 💾</button>

</div>

</div>

</div>

<!-- ТРИГГЕР-ШЕСТЕРЕНКА ВХОДА В ИНЖЕНЕРНЫЙ РЕЖИМ -->

<div style="text-align: center; margin-top: 16px; margin-bottom: 24px;">

⚙️


```
</div>
```

```
</div> <!-- Конец container -->
```

```
<!-- МОДАЛЬНОЕ ОКНО АУТЕНТИФИКАЦИИ MUSHROOM -->
```

```
<div id="authModal" style="display:none; position:fixed; top:0; left:0; width:100%; height:100%; background:rgba(0,0,0,0.6); backdrop-filter:blur(8px); z-index:9999; align-items:center; justify-content:center;">
```

```
  <div class="m-card" style="width:90%; max-width:340px; padding:24px; text-align:center; animation: slideUp 0.3s ease-out;">
```

```
    <div style="font-size:40px;"><img alt="lock icon" data-bbox="290 260 310 275"/></div>
```

```
    <div class="card-value" style="font-size:20px; margin-bottom:6px;">Вход в инженерию</div>
```

```
    <div class="card-label" style="margin-bottom:20px;">Введите пароль администратора</div>
```

```
    <input type="password" id="web_password_input" placeholder="Введите пароль" style="text-align:center; margin-bottom:16px; font-size:16px; padding:12px;" onkeypress="if(event.key === 'Enter') submitAuth()">
```

```
    <div style="display:flex; gap:10px;">
```

```
      <button class="secondary" onclick="closeAuthModal()">Отмена</button>
```

```
      <button class="success" onclick="submitAuth()">Войти</button>
```

```
    </div>
```

```
</div>
```

```
</div>
```

```
<script src="/uPlot.iife.min.js"></script>
```

```
<script>
```

```
  let tempChart, pressureChart;
```

```
  let settingsOpen = false;
```

```
  let scanActive = false;
```

```
  let scanResults = new Array(21).fill(-120);
```

```
  let scanCurrentStep = 0;
```

```
  let scanLiveRssi = -120;
```

```
  let scanInterval = null;
```

```
  let sessionToken = localStorage.getItem('gate_session_token') || "";
```

```
  let userRole = localStorage.getItem('gate_user_role') || "";
```

```
  let pendingAction = null;
```

```
  const windDirections16 = ["C", "CCB", "CB", "BCB", "B", "БЮБ", "ЮБ", "ЮЮБ", "Ю", "ЮЮЗ", "ЮЗ", "ЗЮЗ", "З", "ЗСЗ", "СЗ", "ССЗ"];
```

```
  let maxTemp = 50, minTemp = -30, minPress = 970, maxPress = 1050, maxRain = 100;
```

```
  function getWindDirection16(deg) { return windDirections16[Math.round(deg / 22.5) % 16]; }
```

```
  function drawCompassRose() {
```

```
    const container = document.getElementById('compassBg');
```

```
    if (!container) return;
```

```

container.innerHTML = "";
const size = 140, centerX = size / 2, centerY = size / 2, radius = size / 2 - 5;
for (let i = 0; i < 16; i++) {
  const angle = (i * 22.5 - 90) * Math.PI / 180;
  const x1 = centerX + radius * 0.85 * Math.cos(angle), y1 = centerY + radius * 0.85 * Math.sin(angle);
  const x2 = centerX + radius * Math.cos(angle), y2 = centerY + radius * Math.sin(angle);
  const isCardinal = (i % 4 === 0), isHalf = (i % 2 === 0);
  const line = document.createElement('div');
  line.style.position = 'absolute'; line.style.left = x1 + 'px'; line.style.top = y1 + 'px';
  line.style.width = Math.hypot(x2 - x1, y2 - y1) + 'px'; line.style.height = isCardinal ? '2px' : '1px';
  line.style.backgroundColor = isCardinal ? 'var(--wind)' : 'var(--text-secondary)';
  line.style.transformOrigin = '0 0'; line.style.transform = `rotate(${i * 22.5}deg)`; line.style.opacity = isHalf ? 0.6 : 0.3;
  container.appendChild(line);
}
[
  { d: 'C', a: -90 }, { d: 'B', a: 0 }, { d: 'IO', a: 90 }, { d: '3', a: 180 }
].forEach(l => {
  const rad = l.a * Math.PI / 180;
  const div = document.createElement('div'); div.className = 'compass-mark';
  div.style.left = (centerX + (radius + 12) * Math.cos(rad)) + 'px'; div.style.top = (centerY + (radius + 12) * Math.sin(rad)) +
'px';
  div.style.fontWeight = 'bold'; div.style.color = 'var(--wind)'; div.innerText = l.d; container.appendChild(div);
});
}

function initTheme() {
  const saved = localStorage.getItem('darkMode');
  if (saved === 'true') document.body.classList.add('dark');
  else if (saved === 'false') document.body.classList.remove('dark');
  setTimeout(() => drawCompassRose(), 100);
}

function toggleTheme() {
  document.body.classList.toggle('dark');
  localStorage.setItem('darkMode', document.body.classList.contains('dark'));
  setTimeout(() => drawCompassRose(), 100);
  if (tempChart) tempChart.destroy(); if (pressureChart) pressureChart.destroy();
  initCharts(); loadHistory(); drawSpectrum();
}

function initCharts() {
  const commonOpts = {
    width: document.getElementById('temp-chart')?.offsetWidth || 400, height: 140,
    series: [{}], axes: [{ show: false }, { show: true, grid: { show: false }, stroke: "var(--text-secondary)" }],
    cursor: { drag: { setScale: false } }
  };
};

```

```

    if (document.getElementById('temp-chart')) {
        tempChart = new uPlot({...commonOpts, series: [{ stroke: "#ff6b35", fill: "rgba(255,107,53,0.06)", spanGaps: false }]},
        [], [], document.getElementById('temp-chart'));
    }
    if (document.getElementById('pressure-chart')) {
        pressureChart = new uPlot({...commonOpts, series: [{ stroke: "#4caf50", fill: "rgba(76,175,80,0.06)", spanGaps:
false }]}, [], [], document.getElementById('pressure-chart'));
    }
}

```

```

function prepareData(values, stepSeconds) {
    const now = Math.floor(Date.now() / 1000); const timestamps = [];
    for (let i = values.length - 1; i >= 0; i--) timestamps.push(now - (values.length - 1 - i) * stepSeconds);
    return [timestamps, values];
}

function formatUptime(s) { let d = Math.floor(s/86400), h = Math.floor((s%86400)/3600), m = Math.floor((s%3600)/60);
return d>0 ? `${d}д ${h}ч` : `${h}ч ${m}м`; }

function updateProgress(v, mn, mx, id) { const f = document.getElementById(id); if (f) f.style.width =
Math.min(Math.max((v - mn) / (mx - mn) * 100, 0), 100) + '%'; }

```

```

function getForecastIcon(txt) {
    if (!txt) return "🕒"; const t = txt.toLowerCase();
    if (t.includes('ясн') || t.includes('солнечн') || t.includes('отличн')) return "☀️";
    if (t.includes('облачн') || t.includes('переменн')) return "🌤️";
    if (t.includes('дожд') || t.includes('ливн')) return "🌧️";
    if (t.includes('гроз') || t.includes('шторм')) return "⚡️";
    return "☁️";
}

```

// СЕССИОННЫЙ СЛОЙ БЕЗОПАСНОСТИ MUSHROOM ДЛЯ ШЕСТЕРЕНКИ

```

function handleGearClick() {
    if (userRole === "admin" && sessionToken) {
        // Если уже вошли, то клик по шестеренке разлогинивает систему (для безопасности)
        sessionToken = ""; userRole = "";
        localStorage.removeItem('gate_session_token'); localStorage.removeItem('gate_user_role');
        applyRoleUI(); alert("Вы вышли из инженерного режима.");
    } else {
        showAuthModal(() => toggleSettings());
    }
}

```

```

function showAuthModal(onSuccessAction) { pendingAction = onSuccessAction;
document.getElementById('authModal').style.display = 'flex'; document.getElementById('web_password_input').focus(); }
function closeAuthModal() { document.getElementById('authModal').style.display = 'none'; pendingAction = null; }

```

```
function applyRoleUI() {
  const isAdmin = (userRole === 'admin');
  document.getElementById('settingsPanelBlock').style.display = isAdmin ? 'block' : 'none';
  document.getElementById('startScanBtn').style.display = isAdmin ? 'block' : 'none';
  document.getElementById('gear_btn').style.opacity = isAdmin ? "1" : "0.5";
}
```

```
function sha256(string) { return CryptoJS.SHA256(string).toString(CryptoJS.enc.Hex).toLowerCase(); }
```

```
async function submitAuth() {
  const pass = document.getElementById('web_password_input').value;
  if (!pass) return;
  const hash = sha256(pass);
  try {
    const res = await fetch('/api/auth', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ password_hash: hash })
    });
    if (res.status === 200) {
      const data = await res.json();
      sessionToken = data.token; userRole = data.role;
      localStorage.setItem('gate_session_token', sessionToken); localStorage.setItem('gate_user_role', userRole);
      applyRoleUI(); closeAuthModal();
      document.getElementById('web_password_input').value = '';
      load();
      if (pendingAction) pendingAction();
    } else { alert('Неверный PIN-код инженера!'); document.getElementById('web_password_input').value = ''; }
  } catch(e) { alert('Ошибка сети'); }
}
```

```
async function secureFetch(url, options = {}) {
  if (!options.headers) options.headers = {};
  options.headers['Authorization'] = `Bearer ${sessionToken}`;
  let response = await fetch(url, options);
  if (response.status === 401) { showAuthModal(() => secureFetch(url, options)); throw new Error("Unauthorized"); }
  return response;
}
```

```
function toggleSettings() {
  const content = document.getElementById('settingsContent'); const arr = document.getElementById('set_arrow');
  content.classList.toggle('open'); settingsOpen = content.classList.contains('open');
  arr.innerText = settingsOpen ? "▲" : "▼";
}
```

```
}
```

```
// РЕАКТИВНЫЙ ОБНОВЛЯЕМЫЙ МОНИТОРИНГ
```

```
async function load() {
```

```
  try {
```

```
    const response = await fetch('/api/data'); const d = await response.json();
```

```
    document.getElementById('temperature').innerText = d.temperature.toFixed(1);
```

```
    document.getElementById('temp_min').innerText = d.temp_min?.toFixed(1) || '--';
```

```
    document.getElementById('temp_max').innerText = d.temp_max?.toFixed(1) || '--';
```

```
    updateProgress(d.temperature, minTemp, maxTemp, 'temp-fill');
```

```
    document.getElementById('humidity').innerText = d.humidity.toFixed(0);
```

```
    updateProgress(d.humidity, 0, 100, 'hum-fill');
```

```
    document.getElementById('pressure').innerText = d.pressure.toFixed(0);
```

```
    document.getElementById('pressure_min').innerText = d.pressure_min?.toFixed(0) || '--';
```

```
    document.getElementById('pressure_max').innerText = d.pressure_max?.toFixed(0) || '--';
```

```
    updateProgress(d.pressure, minPress, maxPress, 'pressure-fill');
```

```
    document.getElementById('wind_speed').innerText = d.wind_speed.toFixed(1);
```

```
    document.getElementById('wind_gust').innerText = d.wind_gust?.toFixed(1) || '--';
```

```
    document.getElementById('wind_max').innerText = d.wind_max?.toFixed(1) || '--';
```

```
    updateProgress(d.wind_speed, 0, 40, 'wind-fill');
```

```
    document.getElementById('wind_dir_deg').innerText = d.wind_direction + '°';
```

```
    document.getElementById('wind_dir_txt').innerText = getWindDirection16(d.wind_direction);
```

```
    const arrow = document.getElementById('windArrow');
```

```
    if (arrow) arrow.style.transform = `rotate(${d.wind_direction}deg) translateX(-50%)`;
```

```
        document.getElementById('windDirLabel').innerHTML = `${d.wind_direction}°<br>${
```

```
{getWindDirection16(d.wind_direction)}}`;
```

```
    document.getElementById('rain').innerText = d.rain.toFixed(1);
```

```
    document.getElementById('rain_max').innerText = d.rain_max?.toFixed(1) || '--';
```

```
    updateProgress(d.rain, 0, maxRain, 'rain-fill');
```

```
    document.getElementById('power_source').innerText = d.power_source === 'AC' ? 'Сеть' : 'АКБ';
```

```
        document.getElementById('mqtt_status').innerHTML = d.mqtt_status === 'OK' ? '<span class="badge badge-success">OK</span>' : '<span class="badge badge-error">Offline</span>';
```

```
    document.getElementById('version').innerText = d.version || '--';
```

```
    document.getElementById('uptime').innerText = formatUptime(d.uptime || 0);
```

```
    document.getElementById('radio_success_header').innerText = d.radio_success || 0;
```

```
    document.getElementById('radio_errors_header').innerText = d.radio_errors || 0;
```

```
    document.getElementById('datetime').innerText = d.datetime || '--';
```

```
    document.getElementById('sensor_id').innerText = d.sensor_id || '--';
```

```

    document.getElementById('outdoor_battery').innerHTML = d.outdoor_battery === 'OK' ? '<span class="badge badge-success">OK</span>' : '<span class="badge badge-error">LOW</span>';
    document.getElementById('current_freq').innerText = d.current_freq?.toFixed(2) || '915.00';
    document.getElementById('bmp_temp_val').innerText = d.bmp_temperature?.toFixed(1) || '0.0';

    const rs = document.getElementById('radio_status');
    if (rs) {
        if (d.radio_errors === 0 && d.radio_success === 0) { rs.innerText = 'НЕТ ДАННЫХ'; rs.className = 'badge badge-warning'; }
        else if (d.radio_errors > 0 && d.radio_success === 0) { rs.innerText = 'ОШИБКИ'; rs.className = 'badge badge-error'; }
    }

    else { rs.innerText = 'OK'; rs.className = 'badge badge-success'; }
}

if (d.forecast) { document.getElementById('forecast_text').innerHTML = d.forecast;
document.getElementById('forecast_icon').innerHTML = getForecastIcon(d.forecast); }
if (d.pressure_trend) document.getElementById('pressure_trend').innerHTML = d.pressure_trend;
if (!scanActive && d.rssi) document.getElementById('scan_rssi').innerText = d.rssi;

if (!settingsOpen && userRole === 'admin') {
    document.getElementById('hostname').value = d.hostname || 'weather_gate';
    document.getElementById('mq_ip').value = d.mq_ip || '';
    document.getElementById('mq_u').value = d.mq_user || '';
    document.getElementById('mq_p').value = d.mqtt_password || '';
    document.getElementById('altitude_input').value = d.altitude || 150;
    document.getElementById('bmp_offset').value = d.bmp_offset || 31.7;
    document.getElementById('mqtt_topic_weather').value = d.mqtt_topic_weather || 'weather_gate/data';
}
} catch(e) { console.error(e); }
}

async function loadHistory() {
    try {
        const response = await fetch('/api/history'); const d = await response.json();
        if (d.temp_history && tempChart) tempChart.setData(prepareData(d.temp_history, 300));
        if (d.pressure_history && pressureChart) pressureChart.setData(prepareData(d.pressure_history, 300));
    } catch(e) { console.error(e); }
}

async function syncTime() {
    if (!confirm('Синхронизировать время с вашим браузером?')) return;
    const now = Math.floor(Date.now() / 1000);
    await fetch('/api/sync_time', { method: 'POST', headers: {'Content-Type': 'application/x-www-form-urlencoded'}, body:
't=' + now });
    alert('Время шлюза синхронизировано!'); load();
}

```

```

}

async function resetRadioErrors() {
  if (confirm('Сбросить счётчики ошибок и успехов шины CC1101?')) {
    await secureFetch('/api/reset_errors', { method: 'POST' }); alert('Статистика сброшена'); load();
  }
}

async function save() {
  if (!confirm('Сохранить инженерные калибровки и перезагрузить шлюз?')) return;
  const obj = {
    hostname: document.getElementById('hostname').value,
    mq_ip: document.getElementById('mq_ip').value,
    mq_u: document.getElementById('mq_u').value,
    mq_p: document.getElementById('mq_p').value,
    altitude: parseInt(document.getElementById('altitude_input').value),
    bmp_offset: document.getElementById('bmp_offset').value,
    mqtt_topic_weather: document.getElementById('mqtt_topic_weather').value
  };
  const np = document.getElementById('web_password').value;
  if (np.length >= 4) obj.new_password = sha256(np);

  await secureFetch('/api/config', { method: 'POST', headers: { 'Content-Type': 'application/json' }, body: JSON.stringify(obj)
});
  location.reload();
}

// СКАНЕР АЧХ И ОТПИСОВКА CANVAS
async function startFullScan() {
  if (scanActive) { alert("Сканирование эфира уже выполняется!"); return; }
  if (!confirm("Запустить автопоиск оптимальной частоты Fine Offset?\nЭто займёт около 23 минут. Приём данных временно приостановится.")) return;
  document.getElementById('startScanBtn').disabled = true; scanActive = true;
  try {
    await secureFetch('/api/start_scan');
    if (scanInterval) clearInterval(scanInterval);
    scanInterval = setInterval(pollScanData, 2000);
  } catch(e) { document.getElementById('startScanBtn').disabled = false; scanActive = false; }
}

async function pollScanData() {
  try {
    const response = await fetch('/api/scan_data'); const data = await response.json();
    scanResults = data.rssi; scanCurrentStep = data.step; scanActive = data.active;
  }
}

```

```

scanLiveRssi = data.instant || data.rssi_current;
drawSpectrum();
if (data.active) {
    const progress = Math.round((data.step / 21) * 100);
    document.getElementById('startScanBtn').innerHTML = ` ⌚ Сканирование: ${progress}%`;
} else {
    if (scanInterval) clearInterval(scanInterval); scanInterval = null;
    document.getElementById('startScanBtn').disabled = false; document.getElementById('startScanBtn').innerHTML =
    ' 🔍 Найти лучшую частоту';
    const dResp = await fetch('/api/data'); const dd = await dResp.json();
    if (dd.rssi) document.getElementById('scan_rssi').innerText = dd.rssi;
}
} catch(e) { console.error(e); }
}

```

```

function drawSpectrum() {
    const canvas = document.getElementById('spectrumCanvas'); if (!canvas) return;
    const w = canvas.offsetWidth, h = canvas.offsetHeight; canvas.width = w; canvas.height = h;
    const ctx = canvas.getContext('2d'); ctx.clearRect(0, 0, w, h);
    ctx.fillStyle = getComputedStyle(document.body).getPropertyValue('--bg').trim(); ctx.fillRect(0, 0, w, h);

    const noiseY = h - ((-108 + 120) * (h / 90));
    ctx.setLineDash([5, 5]); ctx.strokeStyle = '#f44336'; ctx.lineWidth = 1; ctx.beginPath(); ctx.moveTo(0, noiseY);
    ctx.lineTo(w, noiseY); ctx.stroke();
    ctx.setLineDash([]); ctx.fillStyle = '#f44336'; ctx.font = '9px sans-serif'; ctx.fillText('ШУМ', 5, noiseY - 3);

    const stepX = w / (scanResults.length - 1);
    ctx.beginPath(); ctx.lineWidth = 2.5; ctx.lineJoin = 'round'; ctx.strokeStyle = scanActive ? '#2196F3' : '#4CAF50';
    let firstPoint = true;
    for (let i = 0; i < scanResults.length; i++) {
        let val = scanResults[i]; if (val === null || val === -120) continue;
        let y = h - ((Math.max(-120, Math.min(-30, val)) + 120) * (h / 90)), x = i * stepX;
        if (firstPoint) { ctx.moveTo(x, y); firstPoint = false; } else ctx.lineTo(x, y);
    }
    ctx.stroke();

    ctx.fillStyle = '#2196F3';
    for (let i = 0; i < scanResults.length; i++) {
        let val = scanResults[i]; if (val === null || val === -120) continue;
        let y = h - ((Math.max(-120, Math.min(-30, val)) + 120) * (h / 90)), x = i * stepX;
        ctx.beginPath(); ctx.arc(x, y, 2, 0, Math.PI * 2); ctx.fill();
    }

    ctx.fillStyle = '#9e9e9e'; ctx.font = '9px sans-serif'; ctx.textAlign = 'center';

```



```

[[{f: 914.80, l: '914.8'}, {f: 915.00, l: '915.0'}, {f: 915.20, l: '915.2'}]].forEach(m => {
  const x = ((m.f - 914.80) / 0.4) * w;
  if (x >= 0 && x <= w) { ctx.beginPath(); ctx.moveTo(x, h); ctx.lineTo(x, h - 6); ctx.stroke(); ctx.fillText(m.l + ' MHz', x, h -
10); }
});

const curFreq = parseFloat(document.getElementById('current_freq').innerText) || 915.00;
const markerX = ((curFreq - 914.80) / 0.4) * w;
if (markerX >= 0 && markerX <= w) {
  ctx.setLineDash([5, 5]); ctx.strokeStyle = '#FF9800'; ctx.lineWidth = 2; ctx.beginPath(); ctx.moveTo(markerX, 0);
ctx.lineTo(markerX, h); ctx.stroke();
  ctx.setLineDash([]); ctx.fillStyle = '#FF9800'; ctx.beginPath(); ctx.arc(markerX, 8, 5, 0, Math.PI * 2); ctx.fill();
  ctx.fillStyle = '#fff'; ctx.font = 'bold 8px sans-serif'; ctx.fillText(curFreq.toFixed(2), markerX, 12);
}
}

function setupSpectrumClick() {
  const canvas = document.getElementById('spectrumCanvas'); if (!canvas) return;
  canvas.onclick = async function(e) {
    if (userRole !== "admin") return; // Ограничение: клик по частоте доступен только инженеру
    if (scanActive) { alert("Сканирование активно, подождите завершения"); return; }
    const rect = canvas.getBoundingClientRect(); const x = e.clientX - rect.left; const w = canvas.offsetWidth;
    const clickedFreq = (914.80 + (x / w) * 0.4).toFixed(3);
    if (confirm(`Настроить гетеродин приёмника на частоту ${clickedFreq} MHz?`)) {
      await secureFetch('/api/set_freq', { method: 'POST', headers: {'Content-Type': 'application/x-www-form-urlencoded'}, body: 'f=' + clickedFreq });
      document.getElementById('current_freq').innerText = clickedFreq; drawSpectrum();
    }
  };
}

initTheme(); initCharts(); setupSpectrumClick(); drawSpectrum(); drawCompassRose();
document.addEventListener('DOMContentLoaded', function() {
  applyRoleUI(); load(); loadHistory();
  setInterval(load, 4000); setInterval(loadHistory, 60000);
  window.onresize = () => {
    if (tempChart) tempChart.setSize({width: document.getElementById('temp-chart').offsetWidth, height: 140});
    if (pressureChart) pressureChart.setSize({width: document.getElementById('pressure-chart').offsetWidth, height:
140});
    drawSpectrum();
  };
});
</script>
</body>
</html>

```